

THE EXACT NON-CENTRAL DISTRIBUTION OF A MULTIVARIATE COMPLEX QUADRATIC FORM OF COMPLEX NORMAL VARIATES

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Summary: The p.d.f., moments and joint p.d.f. of the
characteristic roots of a multivariate complex
quadratic form of complex normal variates are
derived. Some definitions and results useful
for the derivation of the distributional results
of the multivariate complex quadratic form are also
considered.

1. INTRODUCTION

In this paper the p.d.f., moments and joint distributions of
the characteristic roots of the quadratic form $\underline{Z}:p \times p = \underline{Z} L \bar{\underline{Z}}'$
are considered. The matrix $L:n \times n$ is a hermitian positive
definite (h.p.d.) matrix and the columns of $\underline{Z}:p \times n = [\underline{Z}_1, \dots, \underline{Z}_n]$
are independently complex normally distributed with common
covariance matrix $\Sigma:p \times p$ and $E(\underline{Z}_j) = \underline{\mu}_j:p \times 1, j=1, \dots, n$.
Let $M:p \times n = [\underline{\mu}_1, \dots, \underline{\mu}_n]$; then the p.d.f. of $\underline{Z}:p \times n$ is

given by

$$f_{\tilde{Z}}(Z) = \pi^{-np} |\Sigma|^{-n} \text{etr}[-\Sigma^{-1}(Z - M)(\overline{Z - M})']$$

with $Z \in \{Z: p \times n = (z_{jk}) : z_{jk} = x_{jk} + iy_{jk}; -\infty < x_{jk}, y_{jk} < \infty, j=1, \dots, p, k=1, \dots, n\}$

and it will be denoted as $\tilde{Z}: p \times n \sim \text{CMTN}(p, n, M, I_n \otimes \Sigma)$.

In order to derive the distributional properties of $\tilde{Z}: p \times p = \tilde{Z} L \tilde{Z}'$, the polynomials $\tilde{P}_\kappa(T, A, B)$, for $T: p \times n$ a complex matrix and $A: n \times n$ and $B: p \times p$ h.p.d. matrices, are defined in section 2 analogous to the polynomials $P_\kappa(T, A, B)$ defined and studied by Crowther (1975) for $A: n \times n$ and $B: p \times p$ real symmetric matrices.

2. THE POLYNOMIALS $\tilde{P}_\kappa(T, A, B)$

Definition 2.1

Let $T: p \times n$ and $W: p \times n, p \leq n$, be arbitrary complex matrices and let $A: n \times n$ and $B: p \times p$ be h.p.d. matrices; then $\tilde{P}_\kappa(T, A, B)$ which corresponds to the partition κ of k is defined as:

$$\begin{aligned} (2.1) \quad \tilde{P}_\kappa(T, A, B) &= \pi^{-np} \text{etr}(T\overline{T}') \int_W \text{etr}\{-W\overline{W}' - i(T\overline{W}' + W\overline{T}')\} \tilde{C}_\kappa(-BWA\overline{W}') dW \\ &= \pi^{-np} \int_W \text{etr}\{-(W+iT)(\overline{W+iT})'\} \tilde{C}_\kappa(-BWA\overline{W}') dW \\ &= \pi^{-np} \int_V \text{etr}\{-V\overline{V}'\} \tilde{C}_\kappa(-B(V-iT)A(\overline{V-iT})') dV \\ &= E[\tilde{C}_\kappa(-B(V-iT)A(\overline{V-iT})')] \end{aligned}$$

where the expectation is with respect to the p.d.f. of $\underline{V}: p \times n$, i.e.

$$(2.2) \quad f_{\underline{V}}(V) = \pi^{-np} \text{Petr}(-V\bar{V}'), \quad V: p \times p > 0$$

$$\text{i.e. } \underline{V}: p \times n \sim \text{CMTN}'(p, n, 0, I_n \otimes I_p).$$

Property 2.1

$$\tilde{p}_k(0, A, B) = (-1)^k [n]_k \frac{\tilde{C}_k(A) \tilde{C}_k(B)}{\tilde{C}_k(I_n)}$$

$$\text{with } [n]_k = \prod_{i=1}^p (n-i+1)_{k_i}.$$

Proof

By using the fundamental property of zonal polynomials of hermitian matrices given by James ((1960), p. 155), (1) of Srivastava ((1965), p. 313-315) and (1.1) of Hayakawa ((1972), p. 232) the result follows.

Property 2.2

$$(2.3) \quad \sum_K \tilde{p}_k(T, A, B) = E[-\text{tr}\{B(V-iT)A(\overline{V-iT})'\}]^k$$

where the expectation is with respect to the p.d.f. in (2.2).

Proof

By using the unique decomposition of $\text{tr}\{B(V-iT)A(\overline{V-iT})'\}$ in terms of zonal polynomials, i.e.

$$\text{tr}\{B(V-iT)A(\overline{V-iT})'\} = \sum_K \tilde{C}_k(B(V-iT)A(\overline{V-iT})'),$$

the result follows.

Lemma 2.1

The j -th cumulant of $\text{tr}\{B(V-iT)A(\overline{V-iT})'\}$, i.e. the coefficient of $\frac{\theta^j}{j!}$ in the series expansion of

$\ln E[\text{etr}\{\theta B(V-iT)A(\overline{V-iT})'\}]$ is given by:

$$(2.4) \quad c_j = (j-1)! \text{tr}(B^j) \text{tr}(A^j) - j! \text{tr}(B^j T A^j \overline{T}')$$

where $\underline{V}: p \times n \sim \text{CMTN}(p, n, 0, I_n \otimes I_p)$.

Proof

By using (5.7) of Crowther ((1973, p. 137) the result follows.

3. DISTRIBUTIONAL RESULTS OF THE QUADRATIC FORM $\underline{S}: p \times p = \underline{Z} \underline{L} \underline{Z}'$

Definition 3.1

The symmetrised p.d.f. of the h.p.d. random matrix $\underline{A}: p \times p$ is defined as

$$(3.1) \quad f_{\text{csym}}(\underline{A}) = \int_{U(p)} f_{\underline{A}}(\underline{U} \underline{A} \underline{U}') d\underline{U}$$

where $f_{\underline{A}}(\underline{A})$ is the p.d.f. of $\underline{A}: p \times p$ and $U(p)$ is the group of $p \times p$ unitary matrices.

Analogous to theorem 2.1 of Greenacre ((1973), p. 97) it can easily be proved that if $g(\underline{A})$ is a function of the random

h.p.d. matrix $\underline{A}: p \times p$ such that $g(\underline{A}) = g(\underline{U} \underline{A} \underline{U}')$, $\underline{U} \in U(p)$; then the distribution of $g(\underline{A})$ is invariant with respect to symmetrisation of the p.d.f. $f_{\underline{A}}(\underline{A})$. This property implies that the p.d.f.'s of $|\underline{A}|$, $|\underline{I}_p - \underline{A}|$, $\underline{D}_{\underline{A}}$ and $\text{tr}(\underline{A})$ and the expected values $E(|\underline{A}|^h)$, $E(|\underline{I}_p - \underline{A}|^h)$ and $E((\text{tr}(\underline{A}))^h)$ are invariant with respect to symmetrisation of the p.d.f. $f_{\underline{A}}(\underline{A})$.

Theorem 3.1

Let $\underline{Z}: p \times n \sim \text{CMTN}(p, n, \underline{M}, \underline{I}_n \otimes \underline{\Sigma})$ and let $\underline{L}: n \times p$ be a h.p.d. matrix; then the p.d.f., symmetrised p.d.f. and moments of $\underline{S}: p \times p = \underline{Z} \underline{L} \underline{Z}'$ and $\underline{D}_{\underline{S}} = \text{Diag}(\tilde{S}_1, \dots, \tilde{S}_p)$ are given below, ($\tilde{S}_1, \dots, \tilde{S}_p$ are the characteristic roots of $\underline{S}: p \times p$):

$$(3.2) f_{\underline{S}}(S) = \frac{\text{etr}[-\underline{\Sigma}^{-1} \underline{M} \underline{M}'] \text{etr}[-q \underline{\Sigma}^{-1} \underline{S}] |\underline{S}|^{n-p}}{\tilde{\Gamma}_p(n) |\underline{L}|^p |\underline{\Sigma}|^n} \frac{\sum_{\kappa} \tilde{P}_{\kappa}(\underline{\Sigma}^{-\frac{1}{2}} \underline{M} (\underline{I}_n - q \underline{L})^{\frac{1}{2}}, \underline{L}^{-1} - q \underline{I}_n, \underline{\Sigma}^{-\frac{1}{2}} \underline{S} \underline{\Sigma}^{-\frac{1}{2}})}{[n]_{\kappa} k!}$$

$$S = \bar{S}' > 0$$

$$\text{where } \tilde{\Gamma}_p(n) = \pi^{\frac{1}{2}p(p-1)} \prod_{i=1}^p \Gamma(n-i+1), \quad q \geq 0.$$

$$(3.3) f_{\text{csym}}(S) = \frac{\text{etr}[-\underline{\Sigma}^{-1} \underline{M} \underline{M}'] |\underline{S}|^{n-p}}{\tilde{\Gamma}_p(n) |\underline{L}|^p |\underline{\Sigma}|^n} \frac{\sum_{\kappa} \tilde{P}_{\kappa}(\underline{\Sigma}^{-\frac{1}{2}} \underline{M}, \underline{L}^{-1}, \underline{\Sigma}^{-1}) \tilde{C}_{\kappa}(S)}{[n]_{\kappa} k! \tilde{C}_{\kappa}(\underline{I}_p)}$$

$$S = \bar{S}' > 0.$$

$$(3.4) f_{\underline{D}_{\underline{S}}}(\underline{D}_{\underline{S}}) = \frac{\pi^{p(p-1)} \text{etr}[-\underline{\Sigma}^{-1} \underline{M} \underline{M}']}{\tilde{\Gamma}_p(n) \tilde{\Gamma}_p(p) |\underline{L}|^p |\underline{\Sigma}|^n} \frac{\sum_{\kappa} \tilde{P}_{\kappa}(\underline{\Sigma}^{-\frac{1}{2}} \underline{M}, \underline{L}^{-1}, \underline{\Sigma}^{-1})}{[n]_{\kappa} k! \tilde{C}_{\kappa}(\underline{I}_p)} x_{[\kappa]}(1)$$

$$\cdot \prod_{i=1}^p \tilde{s}_i^{n-p} |(\tilde{s}_j^{k_i+p-i})| |(\tilde{s}_j^{p-i})|, \quad 0 < \tilde{s}_1, \dots, \tilde{s}_p,$$

where $\tilde{s}_1, \dots, \tilde{s}_p$ are the values of $\tilde{S}_1, \dots, \tilde{S}_p$ and $|(\tilde{s}_j^{k_i+p-i})|$

and $|\tilde{s}_j^{p-i}|$ are determinants with ij -th element $\tilde{s}_j^{k_j+p-i}$ and $\tilde{s}_j^{p-i}$ respectively.

$$(3.5) \quad E(|S|^h) = \frac{\text{etr}[-\Sigma^{-1}\bar{M}\bar{M}'] \tilde{\Gamma}_p^{(n+h)q(n+h)p}}{\tilde{\Gamma}_p(n) |L|^p} \frac{\sum_{\kappa} \frac{[n+h]_{\kappa} \tilde{p}_{\kappa} (\Sigma^{-\frac{1}{2}} M (I_n - qL), (L^{-1} - qI_n) q^{-1})}{[n]_{\kappa} k!}}{\sum_{\kappa} \frac{[n]_{\kappa} \tilde{p}_{\kappa} (\Sigma^{-\frac{1}{2}} M (I_n - qL), (L^{-1} - qI_n) q^{-1})}{[n]_{\kappa} k!}}$$

Proof

(3.2)

$$\begin{aligned} (3.6) \quad f_{\tilde{S}}(S) &= \pi^{-np} |\Sigma|^{-n} \int_{ZL\bar{Z}'=S} \text{etr}[-\Sigma^{-1}(Z-\bar{M})(\bar{Z}-\bar{M})'] dZ \\ &= \pi^{-np} |\Sigma|^{-n} \int_{ZL\bar{Z}'=S} \text{etr}[-\Sigma^{-1}\{qZL\bar{Z}' + Z(I-qL)\bar{Z}' - M\bar{Z}' - Z\bar{M}' + \bar{M}\bar{M}'\}] \\ &\quad \cdot \int_V \pi^{-np} \text{etr}[-\{V-i(\Sigma^{-\frac{1}{2}}Z(I_n-qL)^{\frac{1}{2}} - \Sigma^{-\frac{1}{2}}M(I_n-qL)^{\frac{1}{2}})\} \\ &\quad \cdot \{V-i(\Sigma^{-\frac{1}{2}}Z(I_n-qL)^{\frac{1}{2}} - \Sigma^{-\frac{1}{2}}M(I_n-qL)^{\frac{1}{2}})\}] dV dX \\ &= \pi^{-2np} |\Sigma|^{-n} \text{etr}[-\Sigma^{-1}\bar{M}\bar{M}' + \Sigma^{-\frac{1}{2}}M(I_n-qL)\bar{M}'\Sigma^{-\frac{1}{2}}] \\ &\quad \cdot \int_V \text{etr}[-V\bar{V}' - iV(I_n-qL)^{\frac{1}{2}}\bar{M}'\Sigma^{-\frac{1}{2}} - i\Sigma^{-\frac{1}{2}}M(I_n-qL)^{\frac{1}{2}}\bar{V}'] \cdot \\ &\quad \cdot \int_{ZL\bar{Z}'=S} \text{etr}[-q\Sigma^{-1}ZL\bar{Z}' + iV(I_n-qL)^{\frac{1}{2}}\bar{Z}'\Sigma^{-\frac{1}{2}} + i\Sigma^{-\frac{1}{2}}Z(I_n-qL)^{\frac{1}{2}}\bar{V}'] dZ dV \end{aligned}$$

In the integral with respect to Z make the transformation $Y = ZL^{\frac{1}{2}}$ with $dZ = |L|^{-p} dY$; then this integral equals

$$\begin{aligned} (3.7) \quad &\int_{\bar{Y}\bar{Y}'=S} \text{etr}[-q\Sigma^{-1}Y\bar{Y}'] \text{etr}[iV(I_n-qL)^{\frac{1}{2}}L^{-\frac{1}{2}}\bar{Y}'\Sigma^{-\frac{1}{2}} + i\Sigma^{-\frac{1}{2}}YL^{-\frac{1}{2}}(I_n-qL)^{\frac{1}{2}}\bar{V}'] dY \\ &= \pi^{np} \text{etr}[-q\Sigma^{-1}S] |S|^{n-p} \tilde{F}_1(n; -\Sigma^{-\frac{1}{2}}V(L^{-1}-qL_n)\bar{V}'\Sigma^{-\frac{1}{2}}S) \\ &\quad \cdot (|L|^{-p} \tilde{\Gamma}_p(n))^{-1} \end{aligned}$$

(from (99) in James (1964), p. 489).

Expanding the hypergeometric function in (3.7) and substitution of (3.7) in (3.6) leads to

$$f_{\underline{S}}(S) = \frac{\text{etr}[-\Sigma^{-1} \bar{M} \bar{M}'] \text{etr}[-q \Sigma^{-1} S] |S|^{n-p}}{\tilde{\Gamma}_p(n) |L|^p |\Sigma|^n} \cdot \sum_k \sum_{\kappa} \frac{1}{[n]_{\kappa} k!} \\ \cdot \pi^{-np} \text{etr}[\Sigma^{-\frac{1}{2}} M (I_n - qL) \bar{M}' \Sigma^{-\frac{1}{2}}] \int_V \text{etr}[-V \bar{V}' - i (\Sigma^{-\frac{1}{2}} M (I_n - qL)^{\frac{1}{2}} \bar{V}' \\ + V (I_n - qL)^{\frac{1}{2}} \bar{M}' \Sigma^{-\frac{1}{2}})] \tilde{C}_{\kappa} (-\Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}} V (L^{-1} - qI_n) \bar{V}') dV.$$

The application of (2.1) leads to (3.2).

(3.3)

Let $q=0$ in (3.2); then

the symmetrised p.d.f. of $\underline{S}: p \times p$ follows from (3.1) by using the fundamental property of zonal polynomials of hermitian matrices.

(3.4)

By using (93) of James ((1964), p. 488) it follows from (3.3) that the joint p.d.f. of the characteristic roots of $\underline{S}: p \times p$ can be written as

$$f_{\underline{D}_S}(\underline{D}_S) = \frac{\pi^{p(p-1)} \text{etr}[-\Sigma^{-1} \bar{M} \bar{M}'] |D_S|^{n-p} \prod_{i>j}^p (\tilde{s}_i - \tilde{s}_j)^2}{\tilde{\Gamma}_p(n) \tilde{\Gamma}_p(p) |L|^p |\Sigma|^n} \\ \cdot \sum_k \sum_{\kappa} \frac{\tilde{p}_{\kappa}(\Sigma^{-\frac{1}{2}} M, L^{-1}, \Sigma^{-1})}{[n]_{\kappa} k! \tilde{C}_{\kappa}(I_p)} \tilde{C}_{\kappa}(\underline{D}_S), \quad 0 < \tilde{s}_1 < \dots < \tilde{s}_p < 0.$$

By using (2.1) in Pillai ((1956), p.1106) and (35) and (85) in James (1964), the p.d.f. in (3.4) follows.

(3.5)

$$E(|\underline{S}|^h) = \frac{\text{etr}[-\Sigma^{-1} M \bar{M}']}{\tilde{r}_p(n) |L|^p |\Sigma|^n} I^*$$

where

$$I^* = \int_{S=S' > 0} \text{etr}[-q \Sigma^{-1} S] |S|^{n+h-p} \sum_{\kappa} \frac{\tilde{P}_{\kappa}(\Sigma^{-\frac{1}{2}} M (I_n - qL)^{\frac{1}{2}}, L^{-1} - qI_n, \Sigma^{-\frac{1}{2}} S \Sigma^{-\frac{1}{2}})}{[n]_{\kappa} k!} dS.$$

By using (2.1) and changing the order of integration the result follows after the application of (1.1) of Hayakawa ((1972), p. 232).

4. CALCULATION OF THE POLYNOMIALS $\tilde{P}_{\kappa}(T, A, B)$

As for the calculations of $P_{\kappa}(T, A, B)$ (cf. Crowther (1975)), the calculations of $\tilde{P}_{\kappa}(T, A, B)$ are also very tedious. In this section expressions for $\tilde{P}_{(1)}(T, A, B)$, $\tilde{P}_{(2)}(T, A, B)$ and $\tilde{P}_{(1,1)}(T, A, B)$ will shortly be derived.

$\tilde{P}_{\kappa}(T, A, B)$ can be written as

$$\tilde{P}_{\kappa}(T, A, B) = \pi^{-np} \int_W \text{etr}\{-(W + iUT)(\overline{W + iUT})' \tilde{C}_{\kappa}(-D_B W A \bar{W}')\} dW$$

where $U: p \times p \in U(p)$ and $D_B = \text{Diag}(\tilde{b}_1, \dots, \tilde{b}_p)$.

From (2.3) follows

$$\sum_{\kappa} \tilde{P}_{\kappa}(T, A, B) = E[\text{tr}(-D_B W A \bar{W}')]]$$

where the expectation is with respect to the p.d.f.

$$\pi^{-np} \text{etr}\{-(W + iUT)(\overline{W + iUT})'\}.$$

From (2.4) follows

$$\begin{aligned}\tilde{P}_{(1)}(T, A, B) &= E[\text{tr}(-D_B W A \bar{W}')] \\ &= -\text{tr}(B) \text{tr}(A) + \text{tr}(A T' B T).\end{aligned}$$

From (2.1) and table 2.2 of Conradie and Troskie ((1982), p. 118) follows that

$$\begin{aligned}(4.1) \quad \tilde{P}_{(2)}(T, A, B) &= E[\tilde{C}_{(2)}(-D_B W A \bar{W}')] \\ &= E\left[\frac{1}{2}((\text{tr}(-D_B W A \bar{W}'))^2 + \text{tr}(-D_B W A \bar{W}')^2)\right] \\ &= \frac{1}{2} E\{\text{tr}(D_B W A \bar{W}')\}^2 + \frac{1}{2} E\{\text{tr}(D_B W A \bar{W}')^2\}\end{aligned}$$

and

$$\begin{aligned}(4.2) \quad \tilde{P}_{(1,1)}(T, A, B) &= E[\tilde{C}_{(1,1)}(-D_B W A \bar{W}')] \\ &= E\left[\frac{1}{2}((\text{tr}(-D_B W A \bar{W}'))^2 - \text{tr}(-D_B W A \bar{W}')^2)\right] \\ &= \frac{1}{2} E\{\text{tr}(D_B W A \bar{W}')\}^2 - \frac{1}{2} E\{\text{tr}(D_B W A \bar{W}')^2\}.\end{aligned}$$

Further it follows that:

$$(4.3) \quad E \text{tr}(D_B W A \bar{W}')^2 = \sum_i \tilde{b}_i^2 E(\underline{W}_i' A \underline{W}_i)^2 + \sum_{i \neq j} \tilde{b}_i \tilde{b}_j E(\underline{W}_i' A \underline{W}_j)^2$$

where $W: p \times n = \begin{bmatrix} \underline{W}_1' \\ \vdots \\ \underline{W}_p' \end{bmatrix}$. Let $U: p \times p = \begin{bmatrix} \underline{U}_1' \\ \vdots \\ \underline{U}_p' \end{bmatrix}$; then from (5.2)

in Crowther ((1973), p. 134) follows

$$E(\underline{W}_i' A \underline{W}_i) = \text{tr}(A) - \underline{U}_i' T A T' \underline{U}_i$$

$$\begin{aligned}
 (4.4) \quad E(\underline{W}_i^t A \underline{\bar{W}}_i)^2 &= \text{the 2-nd cumulant of } \underline{W}_i^t A \underline{\bar{W}}_i + \{E(\underline{W}_i^t A \underline{\bar{W}}_i)\}^2 \\
 &= \text{tr}(A^2) - 2\underline{U}_i^t T A^2 \bar{T}' \underline{\bar{U}}_i + (\text{tr}(A))^2 - 2 \text{tr}(A) \underline{U}_i^t T A \bar{T}' \underline{\bar{U}}_i \\
 &\quad + (\underline{U}_i^t T A \bar{T}' \underline{\bar{U}}_i)^2
 \end{aligned}$$

and

$$\begin{aligned}
 E(\underline{W}_i^t A \underline{\bar{W}}_j \underline{W}_j^t A \underline{\bar{W}}_i | \underline{W}_j) &= \text{tr}(A \underline{\bar{W}}_j \underline{W}_j^t A) - \underline{U}_i^t T A \underline{\bar{W}}_j \underline{W}_j^t A \bar{T}' \underline{\bar{U}}_i \\
 &= \underline{W}_j^t A^2 \underline{W}_j - \underline{W}_j^t A \bar{T}' \underline{\bar{U}}_i \underline{U}_i^t T A \underline{\bar{W}}_j.
 \end{aligned}$$

Hence

$$\begin{aligned}
 (4.5) \quad E(\underline{W}_i^t A \underline{\bar{W}}_j)^2 &= \text{tr}(A^2) - \underline{U}_i^t T A^2 \bar{T}' \underline{\bar{U}}_j - \text{tr}(A \bar{T}' \underline{\bar{U}}_i \underline{U}_i^t T A) \\
 &\quad + \underline{U}_i^t T A \bar{T}' \underline{\bar{U}}_i \underline{U}_i^t T A \bar{T}' \underline{\bar{U}}_j \\
 &= \text{tr}(A^2) - \underline{U}_j^t T A^2 \bar{T}' \underline{\bar{U}}_j - \underline{U}_i^t T A^2 \bar{T}' \underline{\bar{U}}_i + (\underline{U}_i^t T A^2 \bar{T}' \underline{\bar{U}}_j)^2.
 \end{aligned}$$

Substitution of (4.5) and (4.4) in (4.3) leads to

$$\begin{aligned}
 (4.6) \quad E \text{tr}(D_B W A \bar{W}')^2 &= \frac{1}{2} \text{tr}(A^2) \text{tr}(B^2) + \frac{1}{2} \text{tr}(A^2) (\text{tr} B)^2 + \text{tr}(B^2) (\text{tr}(A))^2 \\
 &\quad + \text{tr}(B T A \bar{T}')^2 - 2 \text{tr}(A) \text{tr}(B^2 T A \bar{T}') \\
 &\quad - \text{tr}(B) \text{tr}(B T A^2 \bar{T}') - \text{tr}(B^2 T A^2 \bar{T}').
 \end{aligned}$$

By using (2.4) it follows that:

$$\begin{aligned}
 (4.7) \quad E\{\text{tr}(D_B W A \bar{W}')\}^2 &= \text{tr}(B^2) \text{tr}(A^2) - 2 \text{tr}(B^2 T A^2 \bar{T}') + (\text{tr}(A))^2 (\text{tr}(B))^2 \\
 &\quad - 2 \text{tr}(B) \text{tr}(A) \text{tr}(A \bar{T}' B T) + (\text{tr}(A \bar{T}' B T))^2.
 \end{aligned}$$

Substitution of (4.6) and (4.7) in (4.1) and (4.2) leads to expressions for $\tilde{P}_{(2)}(T, A, B)$ and $\tilde{P}_{(1,1)}(T, A, B)$ which can be calculated.

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